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B.E. (Computer) Semester-VII (Revised Course 2019-20)**EXAMINATION JANUARY 2023****Compiler Design****[Time: Three Hours]****[Max. Marks:100]****Instructions:**

- 1) Attempt Any Two Questions from PART-A and PART-B and One Question from PART-C
- 2) Make Suitable Assumptions Whenever Necessary

PART-A

Q1 a) Consider the assembly program given below.

[10]

```

START 200
MOVER AREG, ='5'
MOVEM AREG, A
LOOP  MULT AREG, A
      MOVER CREG, B
      ADD CREG, ='1'
      BC ANY, AGAIN
      LTORG
      ='5'
      ='1'
AGAIN  SUB AREG, ='1'
      BC LT, BACK
LAST   STOP
      ORIGIN LOOP + 4
      MULT CREG, B
      ORIGIN LAST + 1
A      DS      1
BACK   EQU     LOOP
B      DC      5
      END
      ='1'

```

- i. Design an assembler to generate machine language code for the above assembly program
- ii. Show the content of all the data structures generated during lexical analysis phase of assembler design.
- b) Describe the role of lexical analyser. [6]
- c) Construct the DFA for the regular expression $RE = (a/b)^* abb$ [4]

Q2 a) Check whether the following grammar is an LL (1) grammar. [8]

 $S \rightarrow iCtS \mid iCtSeS \mid a$ $C \rightarrow b$

- b) Discuss the features of YACC tool. [4]
- c) Define context free grammar for the following statements. [8]
 - i. Nested for loop with atleast one if statement inside for loop.
 - ii. Nested if statement with more than assignment statement inside if loop

- Q3 a) For the given LEX program, give the DFA implementation of the lexical analyser for the following [8]
regular definitions.

```
% {A, AB, ABB %}
%%
{a} {yylval = 1; return (A);}
{a*b+} {yylval = 2; return (AB);}
{a*b*} {yylval = 3; return (ABB);}
%%
```

- b) Explain with the help of pseudocode the use of the sentinel character in the input buffering. [6]
c) Write a short note on Front end and back end phases of compiler. [2]
d) Eliminate left Recursion from the following grammar. [4]
E->E+T T
T-> T FF
F-> (E) | id

PART-B

- Q4 a) Construct SLR parsing table for the following grammar [8]

```
S-> P
P->R|R
R-> (S) | id
```

- b) Explain the translation to produce three address code for assignment statements. [8]
c) Write a short on the errors that can occur in the front end phase of the compiler. [4]

- Q5 a) Explain the code generation algorithm with the help of an example. [7]

- b) Write a short note on next use information. [3]
c) Consider the following statements.

```
sum = 0.0
for (i=0; i≤ 10; i++)
{
    sum = sum +a[i];
    sum=sum + sum
}
```

- i. Construct the basic block for the above code. [4]
ii. Optimize each of the block using different optimization techniques. [6]

- Q6 a) What is Symbol table? Describe the contents of Symbol table. [6]

- b) Write the algorithm to compute leading and trailing of a grammar. [5]
c) Write a short note on Backpatching. [5]
d) Draw DAG for the following [4]

```
t1 = a * a
t2 = a * b
t3 = 2 * t2
t4 = t1 + t3
t5 = b * b
t6=t4=t5
```

PART-C

- Q7 a) Compile the following set of C statements by showing the input and output of each phase of compilation. [8]
- ```
int x=0;
while (x <5)
{
 x = x+ 1;
}
```
- b) Write a short note on Evolution of Programming Languages. [4]
- c) Explain the role of parser with the help of a diagram in compiler construction. [4]
- d) Explain the features of Recursive Decent Parser. [4]
- Q8 a) Translate the following statement into Quadruples, Triples and Indirect Triple [8]
- $S = x[i] + y[i] * s$
- b) Consider the following grammar [8]
- $E \rightarrow E \text{ or } T \mid T$
- $T \rightarrow T \text{ and } F \mid F$
- $F \rightarrow \text{not } F \mid \text{true} \mid \text{false} \mid (F)$
- Construct operator precedence parsing table
- c) Explain the technique of peephole optimization with the help of an example. [4]